

BEE 4750 Homework 5: Mixed Integer and Stochastic Programming

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Due Date

Thursday, 12/04/24, 9:00pm

Overview

Instructions

- In Problem 1, you will use mixed integer programming to solve a waste load allocation problem.
- In Problem 2, you will formulate a stochastic optimization problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
Activating project at `~/hw5-ero23_hw5`
```

```

using JuMP
using HiGHS
using DataFrames
using GraphRecipes
using Plots
using Measures
using MarkdownTables

```

Problems (Total: 30 Points)

Problem 1 (24 points)

Three cities are developing a coordinated municipal solid waste (MSW) disposal plan. Three disposal alternatives are being considered: a landfill (LF), a materials recycling facility (MRF), and a waste-to-energy facility (WTE). The capacities of these facilities and the fees for operation and disposal are provided below.

- **LF:** Capacity 200 Mg, fixed cost \$2000/day, tipping cost \$50/Mg;
- **MRF:** Capacity 350 Mg, fixed cost \$1500/day, tipping cost \$7/Mg, recycling cost \$40/Mg recycled;
- **WTE:** Capacity 210 Mg, fixed cost \$2500/day, tipping cost \$60/Mg;

The MRF recycling rate is 40%, and the ash fraction of non-recycled waste is 16% and of recycled waste is 14%. Transportation costs are \$1.5/Mg-km, and the relative distances between the cities and facilities are provided in the table below.

City/Facility	Landfill (km)	MRF (km)	WTE (km)
1	5	30	15
2	15	25	10
3	13	45	20
LF	-	32	18
MRF	32	-	15
WTE	18	15	-

The fixed costs associated with the disposal options are incurred only if the particular disposal option is implemented. The three cities produce 100, 90, and 120 Mg/day of solid waste, respectively, with the composition provided in the table below.

Component	% of total mass	Combustion ash (%)	MRF Recycling rate (%)
Food Wastes	15	8	0
Paper & Cardboard	40	7	55
Plastics	5	5	15
Textiles	3	10	10
Rubber, Leather	2	15	0
Wood	5	2	30
Yard Wastes	18	2	40
Glass	4	100	60
Ferrous	2	100	75
Aluminum	2	100	80
Other Metal	1	100	50
Miscellaneous	3	70	0

The information in the above table will help you determine the overall recycling and ash fractions. Note that the recycling residuals, which may be sent to either landfill or the WTE, have different ash content than the ash content of the original MSW. You will need to determine these fractions to construct your mass balance constraints.

Reminder: Use `round(x; digits=n)` to report values to the appropriate precision!

Problem 1.1

Based on the information above, calculate the overall recycling and ash fractions for the waste produced by each city.

```
RECYCLE_rate= (.15*0+.4*.55+.05*.15+.03*.1+.02*0+.05*.3+.18*.4+.04*.6+.02*.75+.02*.8+.01*.5+
println("Recycling fraction = ", round(RECYCLE_rate; digits=4), '%')
ASH_rate = (.15*.08+.4*.07+.05*.05+.03*.1+.02*.15+.05*.02+.18*.02+.04*1+.02*1+.02*1+.01*1+.03*
println("Ash fraction = ", round(ASH_rate; digits=4), '%')
```

Recycling fraction = 37.75%

Ash fraction = 16.41%

Problem 1.2

What are the decision variables for your optimization problem? Provide notation and variable meaning.

I am using similar decision variables to what we used in the class example...

W_{ij} = Waste transported from city i to disposal j (Mg/day)

cities = (1,2,3)
 disposal = (MRF, WTE, LF)
 R(1,2,3) = Residual waste transported (MRF-WTE , MRF-LF , WTE-LF) (Mg/day)
 Yj = Operational status (on/off) of disposal j (binary)

Problem 1.3

Formulate the objective function. Make sure to include any needed derivations or justifications for your equation(s).

```
Full_MRF_Tipping = 7+(.3775*40) #accounting for only 37.75% of the Mg being recyclable
println("MRF Tipping Cost= ", round(Full_MRF_Tipping; digits=4), '%')
```

MRF Tipping Cost= 22.1%

broadly the objective is:
 minimize (Wij,Y)

sum(fixed_cost[j]* Y[j] for j in facilities) + sum(tipping[j]* W[i,j] for i in cities, j in facilities)
 + transport * sum(dist[(i,j)]*W[i,j] for i in cities, j in facilities)

Beuase I updated the MRF tipping cost above I don't need to add anything specical about recycling and can just use that value in the tipping cost calculations!

$W_{City, plant}^{(1,2,3)}, (MRF, WTE, LF) \quad R^{(1-3)} \quad (MRF \rightarrow WTE, MRF \rightarrow LF, WTE \rightarrow LF)$

MRF cost $\rightarrow 1500 \cdot y_1 + 22.1(W_{11, MRF} + W_{21, MRF} + W_{31, MRF})$
 (1)

WTE cost $\rightarrow 2500 \cdot y_2 + 60(W_{12, WTE} + W_{22, WTE} + W_{32, WTE} + R_{1, MRF} + R_{2, MRF})$
 (2)

LF cost $\rightarrow 2000 \cdot y_3 + 56(W_{13, LF} + W_{23, LF} + W_{33, LF} + R_{1, MRF} + R_{2, MRF} + R_{3, LF})$
 (3)

transport $\rightarrow 1.5(W_{11} \cdot 30 + W_{12} \cdot 15 + W_{13} \cdot 15 + W_{21} \cdot 25 + W_{22} \cdot 10 + W_{23} \cdot 15 + W_{31} \cdot 45 + W_{32} \cdot 20 + W_{33} \cdot 15 + R_1 \cdot 15 + R_2 \cdot 32 + R_3 \cdot 18)$

$\rightarrow$

min $67.1W_{11} + 59.6W_{21} + 89.6W_{31} + 82.5W_{12} + 75W_{22} + 90W_{32} + 57.5W_{13} + 72.5W_{23} + 69.5W_{33} + 82.5R_1 + 98R_2 + 77R_3 + 1500y_1 + 2500y_2 + 2000y_3$
 W, R, y

constraints: $W_{11} + W_{12} + W_{13} = 100$
 (waste amount...) $W_{21} + W_{22} + W_{23} = 90$
 $W_{31} + W_{32} + W_{33} = 120$
 (WTE $\rightarrow$ LF) $R_3 = .1641(W_{12} + W_{22} + W_{32} + R_1)$
 $R_1 + R_2 = .6225(W_{11} + W_{21} + W_{31} + W_{23})$

MRF: $W_{11} + W_{21} + W_{31} \leq 350$
 WTE: $W_{12} + W_{22} + W_{32} + R_1 \leq 210$
 LF: $W_{13} + W_{23} + W_{33} + R_2 + R_3 \leq 200$

$y_1 = \begin{cases} 0 \\ 1 \end{cases} \quad \text{if } W_{11} + W_{21} + W_{31} = 0$
 $y_2 = \begin{cases} 0 \\ 1 \end{cases} \quad \text{if } W_{12} + W_{22} + W_{32} + R_1 = 0$
 $y_3 = 1 \quad (\text{because some waste sent to landfill no matter what})$

$W_{ij}, R_1, R_2, R_3 \geq 0$

I also included my calculations and the exact objective below in the image

Problem 1.4

Derive all relevant constraints. Make sure to include any needed justifications or derivations.

Problem 1.5

Find the optimal solution (using JuMP to solve the problem). Report the optimal objective value.

Problem 1.6

Draw a diagram showing the flows of waste between the cities and the facilities. Which facilities (if any) will not be used? Does this solution make sense?

Problem 2 (6 points)

Consider a two-period economic dispatch problem, based on the multi-period example from Lecture 14 (on 10/29). The generator data, including ramping constraints for each generator, is provided in ‘data/generators.csv.’ In period 1, the demand is $d_1 = 1100\text{MW}$. In period 2, the demand is projected to be $d_2 = 1200\text{MW}$, but there is a 25% probability that it is \$1500. In the first period, the solar capacity factor is 0.9 and the wind capacity factor is 0.45, but in the second period, there is some uncertainty: the forecasted solar and wind capacity factors are 0.95 and 0.4, respectively, but there is a 30% probability that they are 0.75 and 0.5. Your goal is to identify how to dispatch your generators to minimize the cost of meeting demand.

Problem 2.1

Draw a scenario tree for this problem.

Problem 2.2

Formulate a stochastic linear program for this problem based on your scenario tree from Problem 2.1 and the data in `data/generators.csv`.

References

List any external references consulted, including classmates.